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Activity Space Investigations for Nonresidents Using Mobile Phone Data

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ABSTRACT

The transportation integration of in urban areas is the main direction of urban development. The regions in Shanghai have strong connection with the Yangtze River Delta mega-city group. This paper analyzes the activity space of nonresidents by using mobile phone data. An identification method has been used to identify nonresidents traveled through the Hongqiao transportation hub for air flights or high-speed trains. After identifying active areas of nonresidents, POI data is introduced to evaluate the built environment to quantify activity space by using the factor analysis method. Three activity space indicators have been proposed to describe the activity space. By using a two-step clustering method, nonresidents are classified into four clusters to better describe their activity space. The result shows that the activity space of nonresidents was distributed in a way that has strong relationship with Metro Line 2, which can strongly support the further connections between Shanghai and the Yangtze River Delta mega-city group.

Key words: Mobile phone data; nonresidents behavior; cluster analysis; activity space; transportation hub

INTRODUCTION

As the most prosperous and dynamic economic regions in China, the Yangtze River Delta mega-city group has become one of the six largest city groups in the world. The target of developing transportation integration in the city group is not to build a bridge for commuters among cities but also to support the intimate collaborations among industries. These cooperation and connection requires human communication. So the aim of transport integration in the city group is to offer opportunities for seamless communication among people in different cities. To find out whether these communications are aggregated in spatial areas can offer strong support for transport planning in the city group.

Transportation hubs are the places that play important roles in traffic connections from one city to the other cities in the city group. Hongqiao transportation hub is one of the two large transportation hubs in Shanghai and the most important hub connecting cities in the Yangtze River Delta area. One of the special situation of this hub is that it simultaneously contained a high-speed railway station and an airport. In this paper, we try to identify the regions in Shanghai which have strong connections with cities in the Yangtze River Delta mage-city group using mobile phone data to analyze the activity space of nonresidents traveling through the Hongqiao hub.

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The main objective of this paper is to develop a method that can effectively identify the aggregated spatial areas of nonresidents' activity space. However, directly observing the activity space of all nonresidents would be confusing since they are composed of people with different social statues with different mobility patterns. So we propose three activity space indicators (ASIs) to represent individual activity space from three critical perspectives (i.e., activity intensity, activity radius and built environment). We then introduce a clustering method to split nonresidents into four groups with homogenous mobility patterns in each group. After clustering, it become much clearer to describe the activity space of nonresidents.

In this paper, we introduce mobile phone data in analyzing nonresident behavior and activity. it allows us to fully track the trip chain of nonresidents in both temporal and spatial dimensions, no matter inside or outside of the transportation system. The rationality of mobile phone data in transport analyses was proved in previous research (Fontaine et al. 2004). The existing research mainly focused on two aspects: obtaining traditional transportation parameters (Bar-Gera 2007; Chen et al. 2014; Iqbal et al. 2014; Jiang et al. 2015) and obtaining new attributes that cannot acquire in the past (Calabrese et al. 2013; González et al. 2008; Reades et al. 2009). However, less previous research analyzed the activity patterns of passengers using specific transportation modes or nonresidents traveled through a large transport hub. In our previous research, we proved that mobile phone data can support the extraction of rail transit uses from individual daily activity (Li et al. 2016; Li et al. 2015), so in this paper we try to extract the daily activity of nonresidents travel through the large transport hub.

DATA SETS

Mobile phone data

The mobile phone data were collected by 23,918 base transceiver stations (BTS) in Shanghai during October, 2011. Each of these BTSs covers an area with the radius various from 500 to 800 meters. The distribution of these BTSs were high-density in the city center area but sparse in the suburb area. This dataset covers 17.5 million cellphones each day, with their location reported approximately once every half an hour. For the privacy policy, the mobile phone numbers of these records have been one-way encrypted into mobile phone identifiers (CELLUARID). The current position of a mobile user can be approximately represent by the position of the BTS that his or her mobile phone is connecting to. The Data Schema of mobile phone data is as follows (see Table 1):

Table 1. Data Schema of Cellphone Records

Field name	Description
CELLUARID	Encoded number of every mobile phone
DATETIME	Exact time of the record
EVENTID	Identifier of the events generating the connection
LAC	Location area code of the BTS
CI	Cell ID of the BTS
LON	Longitude of the BTS
LAT	Latitude of the BTS

POI Data

With the advancement of intelligent mobile devices, a new type of social web application

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called Location-Based Social Network (LBSNs) services emerged, such as Baidu, Dianping in China. These LBSNs offer us a handy way to search for points of interest (POIs) we interested in and provide reviews or comments on them. With the information of POIs, we can evaluate the built environment in a region.

The POI data are extracted from *Baidu* in 2011, which include totally 775,688 POIs spread all over Shanghai. These POIs have 455 categories distributed in a 4-level hierarchy. The first level distinguishes 21 major categories. The Data Schema of POI data is as follows in Table 2:

Table 2. Data Schema of POIs Information

Field name	Description
NAME	The name of POI
ADDR	The exact address of the POI
CLA	Type of the POI
LON	Longitude of the POI
LAT	Latitude of the POI

NONRESIDENTS IDENTIFICATION

User Segmentation

We screen out the users who have ever shown up in the Hongqiao hub (3.7 million users in total). According to the number of days they showed up in Shanghai in Figure 1, we divided them into three groups:

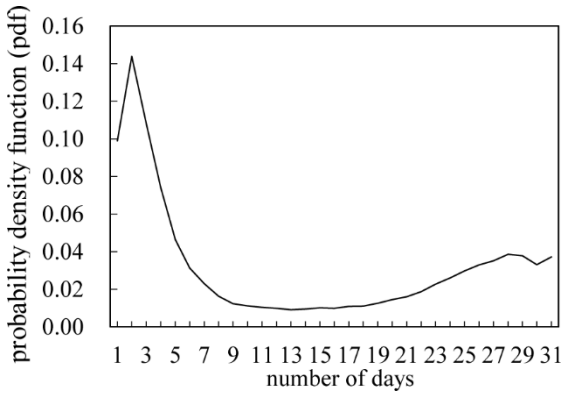

Figure 1. Number of days Hongqiao users shown up in Shanghai

- Users in Group One are those who appeared no more than 9 days, with the amount of about 1.6 million. We consider these users as nonresidents.
- Users in Group Two are those who appeared no less than 10 days but no more than 22 days, which has the amount of about 0.5 million. These users are regarded as the combination of nonresidents and residents.
- Users in Group Three are those who appeared no less than 23 days, with the amount about 1.5 million. This group of users are considered to be the residents of Shanghai who have ever been to the Hongqiao hub in October.

Air passenger identification

Whenever passengers choose airplane as transport mode, they won’t be in Shanghai several

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hours before they arrived and they also won't be in Shanghai several hours after they departed. So the records which are appeared with the patterns listed below are considered as the indication of air arrivals (departures):

- Within t hours before (after) record A, there were no records for the user.
- The BTS that generate record A was in the Hongqiao airport.

By applying the identification method mentioned above, we identify all air travels for three groups to compare the amount with the air records in October, 2012. From our test, setting the threshold to 8 or 20 hours has little influence to the records identified in group one, but has a relatively high influence to group two and group three. In order to avoid mistakenly identified the residents around the Hongqiao hub as air passengers when their phones turn on in the morning or turn off in the evening, the threshold for group one is set as 8 hours and the threshold for both group two and three are set as 20 hours. The results are shown as below in Figure 2):

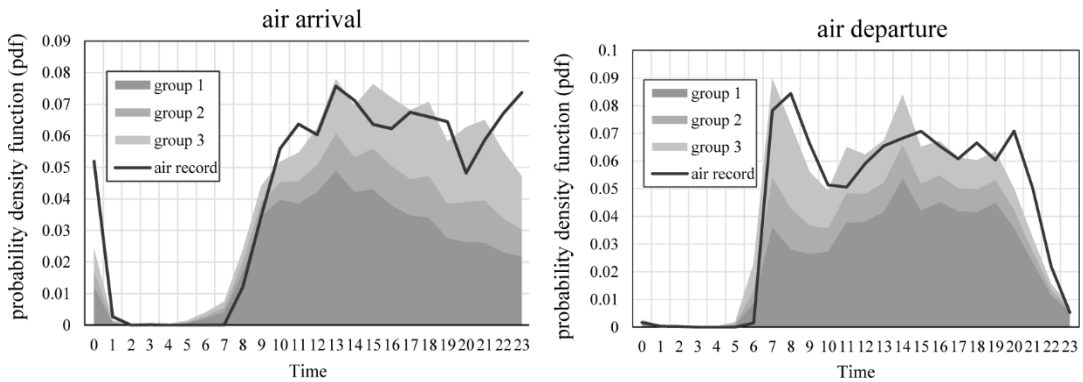

Figure 2. Identified air travels

High-speed rail passenger identification

There are two high-speed rail lines connected to the Hongqiao hub – Beijing-Shanghai Express Railway and Shanghai-Kunming Express Railway. We divide the BTSs within 1.5 kilometers along both rail lines into six sections, with three sections for each rail line.

We consider Record A which is appeared with the patterns listed below as the indication of arrival (departure) of high-speed railway:

- Record A is the first (last) record of a user in the Hongqiao hub during an 8 hours period.
- Within 30 minutes before (after) record A, there exists a Record B, which had no other records during the 8-hour period before (after) it. We consider Record B as the first (last) record of a nonresident entering (leaving) the Shanghai region.
- For the record set φ that between Record A and Record B, it has to contain records that at least generated from two different sections from one rail line.

At the same time, we define Record A and Record B appeared with the pattern listed below as the indication of a user only pass by the Hongqiao hub. These records should be eliminated in the active point identification hereafter this paper. According to the train schedule, user passing by will only stay at the hub less than 30 minutes. So, the identification rules are:

- Record A have been identified as arrival and record B have been identified as departure.
- Record B was generated within 30 minutes after record A.

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- Record A and record B are identified in different rail line.

With the identification method mentioned above, we identified the records of arrival and departure separately and compare them with the number of arrival and departure trains in October (see Figure 3).

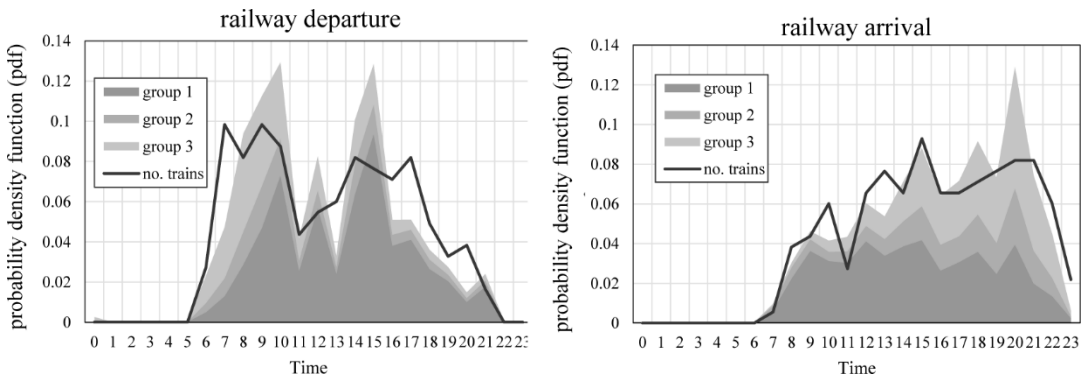

Figure 3. Identified railway travels

DATA PRE-PROCESSING

We extract the raw mobile phone records of the users who have been identified as air or high-speed rail passengers (exclude those who only pass by) in group one (appeared no more than 9 days). These passengers, who we consider as the nonresidents travel through the hongqiao hub, are the main subject of our research.

Active points identification

After eliminate the noise with the method mention in our previous research (Cheng 2015), we can receive a set of control points for each user, formulated as:

$$c_i\left(b_i, t_i\right), i = 1, 2, \ldots \tag{1}$$

where c_i refers to the i-th control point of a user; t_i represent the timestamp of the control point; b_i refers to the BTS of the i-th control point. Given the i-th control point and the following k control points where $b_{i+k} \neq b_{i+k-1} = \ldots = b_i \neq b_{i-1}, k \geq 1$, the activity duration of the user at b_i can be calculated as:

$$t_i^{'} = t_{i+k-1} - t_i \tag{2}$$

where $t_i^{'}$ represent the activity duration. The concept of active point mentioned in this paper is based on the concept of space-time stations (Buscher and Urry 2009) in time geography. In this paper, we define the position that a user stays no less than 30 minutes as an active point. The threshold of 30 minutes is based on the previous survey conducted by our research group in five communities in Shanghai which proves that 30 minutes can represent the critical station of an individual's daily movement and contribute to the comprehensive understanding of individual activities (Cheng 2015; Li et al. 2016).

Built environment assessment

In social science, the term *built environment* refers to the man-made surroundings that provide the setting for human activity (Roof and Oleru 2008). From this definition, a place with

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the better built environment should have more POIs. So we apply POI data to assess the built environment for the activity space of every nonresident. Since the actual size of some of POIs varies from less than one grid to several grids, we process POI data in the traditional traffic analysis zone (TAZ) level to ensure the accuracy.

TABLE 3 Variables description

Variables	Symbol	Description
Resorts density	X_1^k	Number of POIs of corresponding type per unit area in the k th TAZ
Companies density	X_2^k	
Mansions density	X_3^k	
Financial institutions density	X_4^k	
Entertainments density	X_5^k	
Healthcare centers density	X_6^k	
Life services density	X_7^k	
Shops density	X_8^k	
Restaurants density	X_9^k	

However, not all categories of POIs are related to the built environment. So we select 9 (21 in total) major categories of POIs (as shown in Table 3) that are in relation with human activity and calculate the distribution density (the number of POI per unit area) of different categories as variables. Because that numbers of POIs are different in categories, to simply consider the total POI density will emphasis the categories with more POIs. So we employed the factor analysis method to summarize the information from variables by identifying common elements from the correlation between them.

The factor analysis method can be described as follows (Ban et al. 2015): First, the Kaiser-Meyer-Olkin measure of sampling adequacy and Bartlett's test of sphericity are performed to ensure the validation of factor analysis. Then, the main factor analysis is used to analyze all variables. In order to determine the number of factors, we extract the factors with eigenvalues no less than 0.7. The identified factors are summarized to ensure representativeness in developing the final indicators. We used the Oblimin method to rotate the factors and try to make interpretation of each factor.

The factor analysis method extract two factors from the 9 categories of POI. These factors have a power of 84.846% over the total distribution. The two factors and the factor loadings of variables are shown in Table 4. Factor loadings represent the correlation coefficient between variables and factors, which can reflect the importance of variables in each factor. Factor 1 (Z_1) include all the variables with factor loadings of 0.6 or above except resorts density, which can be interpret as the common built environment factor. The variance contribution rate of factor 1 is 76.086%, which means it can explain 76.086% of the information in all variables. Factor 2 (Z_2) only have the variable resort density that have the factor loadings above 0.6 with the variance contribution rate of 8.760%, which can be interpret as the tourism development aspect.

We define the score of each factor and the score of built environment as:

$$Z_i^k = \sum_j L_{ij} X_j^k \tag{3}$$

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$$B_E^k = \frac{\sum_i C_i Z_i^k}{\sum_i C_i} \tag{4}$$

Z_i^k is the score of the i th factor in the k th TAZ;

L_{ij} is the factor loading of the j th variable for the i th factor;

B_E^k is the built environment index of the k th TAZ;

C_i is the variance contribution rate of the i th factor.

Table 4. Result of factor analysis

Variables	Factor loadings	
	Factor 1 (Z_1)	Factor 2 (Z_2)
Resorts density	0.535	0.827
Companies density	0.914	-0.007
Mansions density	0.865	0.140
Financial institutions density	0.910	-0.059
Entertainments density	0.889	-0.151
Healthcare centers density	0.904	-0.184
Life services density	0.944	-0.119
Shops density	0.894	-0.101
Restaurants density	0.922	-0.001

NONRESIDENTS CLUSTERING

Activity Space Indicators

From the active points identification method mentioned above, we can obtain the spatio-temporal activity chain for each nonresident. In order to classify the nonresidents into homogeneous nonresident groups, we propose three activity space indexes (ASIs) which can describe their behavior pattern. These indexes include: daily number of active points (I_A), daily activity range (R_A) and built environment index (B_E). B_E is the average built environment index of one's active points, with the duration as the weight.

Clustering Result

There are totally 915,899 nonresidents who have been identified as air or high-speed rail passengers with at least one active point during the whole month in October, 2011. Since the sample size is so large, we chose the two-step cluster analysis method to classify the nonresidents. This method can automatically define the optimal number of clusters. Therefore, it is ideal for the clustering analysis based on the large-scale dataset obtained through mobile phone data (Li et al. 2016).

The clustering process categorizes the nonresidents into four clusters as shown in Table 5 and in Figure 4. By comparing the index distribution among clusters, we interpret their activity patterns and name them after their characteristics: long distance travel nonresidents (Cluster 1), Low activity intensity nonresidents (Cluster 2), high activity intensity nonresidents (Cluster 3) and frequent visitor of prosperous areas (Cluster 4).

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TABLE 5 Brief summary of the clustering results

Cluster		1	2	3	4
Cluster size (%)		18.6	44.0	19.0	18.4
Cluster centers	I_A	2.02	1.96	4.48	2.65
	R_A (km)	14.03	4.56	5.81	4.68
	B_E	0.11	0.6	1.28	3.49

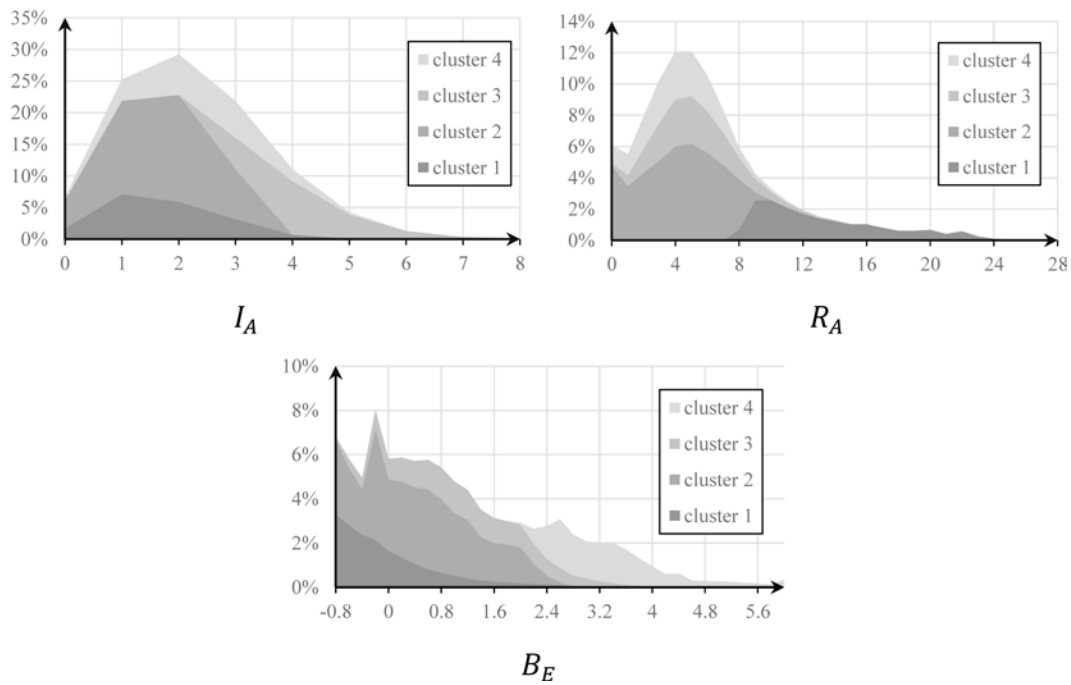

Figure 4. Distribution of ASIs of four clusters

Activity space of four clusters are shown in Figure 5. For Cluster 1, the distribution of active points is more dispersed than others, with a main center in the city center and several small centers spread in the suburban areas. Active space of Cluster 2 distributed in a multi-center shape, with several main centers clustered inside the outer ring road. Although Cluster 3 have the higher activity intensity than Cluster 2, it distributed in a more concentrated way with a main center inside the inner ring road and sporadic active points between the inner ring road and outer ring road. As for Cluster 4, since its built environment index is the highest among all four groups, most of the activities distributed inside the inner ring road on the west side of the Huangpu river, which is the CBD area of Shanghai with the busiest and most affluent streets. Comparing to other clusters, activity areas for this cluster are most concentrated.

By comparing the activity space of all four clusters with Metro Line 2, we can see that the activity space of the four clusters are aggregated in several areas. Most of these areas are distributed along Metro Line 2. This result reflects that the areas along Metro Line 2 are strongly connecting to the cities in the Yangtze River Delta city group. These areas should be significantly considered in traffic planning of the city group.

DISCUSSION AND CONCLUSION

The emergence of big individual tracking data sets brings new opportunities and challenges

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to better understand nonresidents activity space in urban environments. In order to identify the regions in Shanghai which have strong connection with cities in the Yangtze River Delta city group, this paper focused on the activity space of nonresidents travel through the Hongqiao hub. We developed three ASIs to describe the activity space of nonresidents and to classify nonresidents. The classification method we proposed classify nonresidents into four groups with different activity space allowing us to observe the activity space in a much clear way. We summarize and compare the major characteristics of activity space patterns of the four clusters.

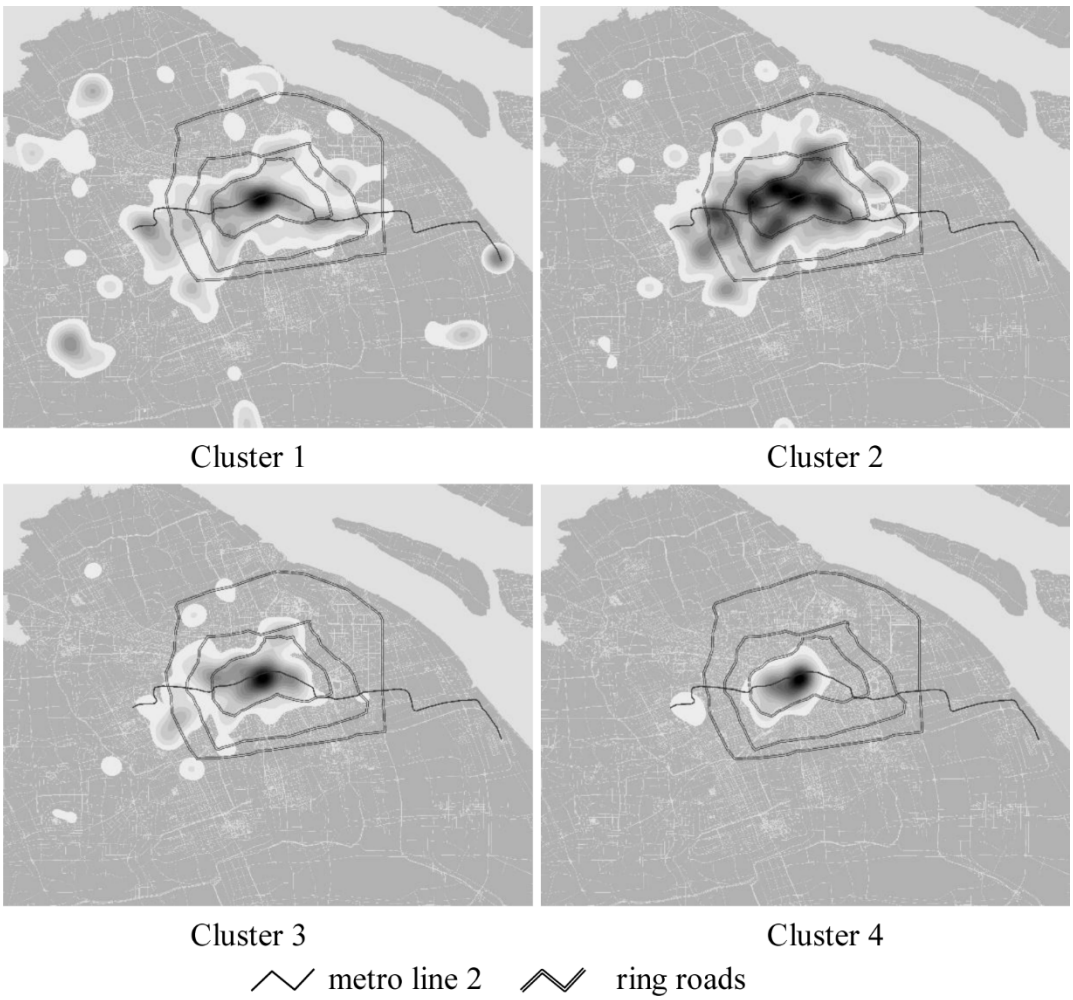

Figure 5. Activity space of four clusters

Our analysis results revealed several interesting aspects of nonresidents activity space in Shanghai. We proposed a method to identify nonresidents. The identified records are distributed similar with the real air and railway records. We proposed three ASIs to help classify nonresidents. These ASIs can help classify nonresidents into groups with different activity space, which allow us to observe the activity space in a much clear way. By observing the activity space of nonresidents, we find that they are aggregated in areas along side with Metro Line 2. This result reflects that human communications inside the city group aggregated in these areas and depend hardly on Metro Line 2. The research findings in this article enhance our understanding of the activity space of nonresidents. We believe that the proposed methods are useful to understand seasonal tourist demand variations and the demands for special events as well as to

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identify the areas closely connected to the other cities.

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